Pfaffian Master Strategy for the Staircase T-System: A Dead-End Report at $n = 4$

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Abstract

The third structural route dreamed on 2026-04-20 for proving the staircase T-system identity $\det M^{(2j+1)}[D_n] \cdot \det M^{(2j-1)}[D_n] = q^{|D_n|}(\det M^{(2j)}[D_n])^2$ uniformly in j was a Pfaffian master matrix in the spirit of Fischer–Gangl. We rule out four natural families of candidates at the smallest test case $n = 4$: (i) the antisymmetrisation Pfaffian $\hat{M}^{(2)} := \begin{pmatrix} 0 & M^{(2)} \\ -M^{(2)\top} & 0 \end{pmatrix}$ gives only the tautology $\mathrm{Pf}(\hat{M}^{(2)})^2 = \det \hat{M}^{(2)} = d_3$, which restates $d_3 = d_2^2$ rather than proving it; (ii) the skew parts $M^{(k)} - M^{(k)\top}$ have either degenerate Pfaffian (for $k = 2$) or Pfaffians with no recognisable factor structure (for $k = 3$); (iii) the Cauchy–Binet expansion of $\det M^{(3)}[D_4]$ obtained from the algebra factorisation $\Pi^{(3)} = \Pi^{(2)} B_3$ has 92 nonzero contributing 6-subsets, none of which dominates; (iv) per-irrep eigenvalue factorisation of $\Pi^{(k)}$ on the regular representation of $H_q(S_4)$ is computable but fails to detect the $(2q+1)^2$ factor of $d_3(4) = q^{18}(q+1)^6(2q+1)^2$, since $(2q+1)$ is *not* an eigenvalue of $\Pi^{(3)}$ on any irreducible S_4-component.

This rules out the simplest structural derivations of the base-case identity at $n = 4$ from a single skew-symmetric master and identifies the missing ingredient: any structural proof must capture cross-irreducible mixing in the D_n-compression that is not visible in the regular-representation spectrum.

1 Setup and the question

We retain the conventions of the companion papers `2026-04-18-first-q-shifted-pair.tex`, `2026-04-19-second-q-shifted-pair.tex`, and `2026-04-20-uniform-locality-reduction.tex`: $H_q(S_n)$ has generators $T_1, \ldots, T_{n-1}$ with $T_i^2 = (q-1)T_i + q$; $\Pi_q^{(k)} := B_1 B_2 \cdots B_k$ where $B_j = (1+T_j)\cdots(1+T_1)$; $D_n := \{w \in S_n : w(2i-1) > w(2i) \text{ for all valid } i\}$, $|D_n| = n!/2^{\lfloor n/2 \rfloor}$; and $M^{(k)}[u,v] = [T_u](T_v \cdot \Pi_q^{(k)})$ on $D_n \times D_n$.

Conjecture. For every $j \geq 1$ and $n \geq 2j+2$,

$$\det M^{(2j+1)}[D_n, D_n] \cdot \det M^{(2j-1)}[D_n, D_n] = q^{|D_n|} \cdot \big(\det M^{(2j)}[D_n, D_n]\big)^2 \quad \text{in } \mathbb{Z}[q]. \tag{1}$$

By uniform locality (companion paper `2026-04-20-uniform-locality-reduction.tex`), (1) for fixed j at all $n \geq K := 2j+2$ reduces to the single base case at $n = K$. The cases $j = 1$ ($K = 4$) and $j = 2$ ($K = 6$) are unconditional (the $K = 4$ base case via direct 6×6 computation; the $K = 6$ base case via degree-bound + multi-evaluation rigorous closure). The $j = 3$ base case at $K = 8$ is open.

The Pfaffian master strategy. A natural structural proof would exhibit a single skew-symmetric matrix $A(q)$ over $\mathbb{Z}[q]$ at each base case $n = K$ such that, on a nested filtration $I_1 \subset I_2 \subset \cdots$ of even-cardinality index sets,

$$\mathrm{Pf}(A|_{I_k}) \;=\; d_k(K), \qquad k = 1, 2, \ldots, K-1,$$

where $d_k(K) := \det M^{(k)}[D_K, D_K]$. The Fischer–Gangl Pfaffian Desnanot–Jacobi (DJ) identity

$$\mathrm{Pf}(A) \cdot \mathrm{Pf}(A_{[ij,kl]}) = \mathrm{Pf}(A_{[ij]})\,\mathrm{Pf}(A_{[kl]}) - \mathrm{Pf}(A_{[ik]})\,\mathrm{Pf}(A_{[jl]}) + \mathrm{Pf}(A_{[il]})\,\mathrm{Pf}(A_{[jk]}) \tag{2}$$

would then deliver (1) as a Plücker-style three-term identity with cancellation of the cross terms.

In what follows we show that the simplest candidates for A at $n = 4$ fail to encode (1), and identify the structural obstruction.

2 Numerical baseline at $n = 4$

By direct computation in SymPy over $\mathbb{Z}[q]$ on $D_4 = \{(2,1,4,3), (3,1,4,2), (3,2,4,1), (4,1,3,2), (4,2,3,1), (4,3,2,1)$

$$\begin{aligned} d_1(4) &= q^6, \\ d_2(4) &= q^9(q+1)^3(2q+1), \\ d_3(4) &= q^{18}(q+1)^6(2q+1)^2 \;=\; \big(d_2(4)\big)^2. \end{aligned}$$

The $j = 1$ identity at $n = 4$ thus takes the perfect-square form $d_3(4) = d_2(4)^2$, equivalently $d_3(4) \cdot d_1(4) = q^6 d_2(4)^2$.

3 Candidate masters and why they fail

3.1 Candidate 1: antisymmetrisation $\hat{M}^{(2)}$

Define

$$\hat{M}^{(2)} := \begin{pmatrix} 0 & M^{(2)}[D_4] \\ -M^{(2)}[D_4]^\top & 0 \end{pmatrix} \;\in\; \mathrm{Mat}_{12}(\mathbb{Z}[q]),$$

which is skew-symmetric.

Observation 1. $\det(\hat{M}^{(2)}) = (\det M^{(2)})^2 = d_2(4)^2 = d_3(4)$, and $\mathrm{Pf}(\hat{M}^{(2)}) = \pm d_2(4)$.

Proof. This is the standard identity for skew block matrices. Computationally verified in SymPy: $\det(\hat{M}^{(2)}) = q^{18}(q+1)^6(2q+1)^2$, and the recursive Pfaffian computation yields $\mathrm{Pf}(\hat{M}^{(2)}) = -q^9(q+1)^3(2q+1)$. □

Remark 2 (Why this is tautological)**.** The identity $\mathrm{Pf}(\hat{M}^{(2)})^2 = (\det M^{(2)})^2$ is a consequence of the construction of $\hat{M}^{(2)}$, not of any property of $M^{(2)}$. To extract (1) as a non-trivial Pfaffian DJ on $\hat{M}^{(2)}$ would require expressing $d_3(4)$ as a Pfaffian of $\hat{M}^{(2)}$ on a different nested submatrix structure — but $d_3(4)$ already equals $\det \hat{M}^{(2)} = \mathrm{Pf}(\hat{M}^{(2)})^2$, so we recover only the perfect-square statement $d_3 = d_2^2$, which is exactly the $j = 1$ identity restated. The Pfaffian DJ (2) on $\hat{M}^{(2)}$ does not produce new information about $d_5, d_7, \ldots$ because $\hat{M}^{(2)}$ has no such sub-structure to exploit.

3.2 Candidate 2: skew parts $M^{(k)} - M^{(k)\top}$

Observation 3. $M^{(2)}[D_4] - M^{(2)}[D_4]^\top$ is singular; $\det = 0$. So no non-degenerate Pfaffian arises here.

Observation 4. $M^{(3)}[D_4] - M^{(3)}[D_4]^\top$ has determinant $q^{12}(q-1)^6(q+1)^2(3q+1)^2(q^4+5q^3+8q^2+6q+3)^2$, which is a square in $\mathbb{Z}[q]$, so the Pfaffian is $\pm q^6(q-1)^3(q+1)(3q+1)(q^4+5q^3+8q^2+6q+3)$.

Remark 5. The factors $(q-1)^3, (3q+1), (q^4+5q^3+8q^2+6q+3)$ do not appear in $d_2(4)$, $d_3(4)$, $d_4(4)$, or $d_5(4)$. They form a parallel family of irreducible polynomials with no apparent relation to the T-system. We do not see how to build a Pfaffian master from $M^{(k)} - M^{(k)\top}$ that recovers the d_k.

3.3 Candidate 3: Cauchy–Binet expansion from algebra factorisation

The factorisation $\Pi^{(3)} = \Pi^{(2)} \cdot B_3$ in $H_q(S_4)$ gives $L_R(\Pi^{(3)}) = L_R(B_3) \cdot L_R(\Pi^{(2)})$ as 24×24 matrices on $H_q(S_4)$ in the T-basis (right-multiplication composes in reverse). Restricting to the $|D_4| = 6$-element index set D_4 on rows and columns and applying Cauchy–Binet:

$$\det M^{(3)}[D_4, D_4] \;=\; \sum_{|J|=6} \det L_R(B_3)[D_4, J] \cdot \det L_R(\Pi^{(2)})[J, D_4], \tag{3}$$

where the sum is over all $\binom{24}{6} = 134{,}596$ size-6 subsets $J \subset S_4$.

Observation 6. At $n = 4$, exactly 92 subsets J contribute non-trivially to (3); their sum reproduces $d_3(4) = q^{18}(q+1)^6(2q+1)^2$. The $J = D_4$ term alone gives $\det L_R(B_3)[D_4, D_4] \cdot d_2(4) = 0 \cdot d_2(4) = 0$ because $L_R(B_3)$ is rank-deficient on D_4. No single contributing term dominates: each contributing minor is a small polynomial of the form $\pm q^a(q+1)^b$ or $\pm q^a(q+1)^b(2q+1)$, and the 92 terms reorganise into the factored form of $d_3(4)$ only after full summation.

Remark 7. A structural proof of $d_3 = d_2^2$ via Cauchy–Binet would require a combinatorial identity collapsing the 92-term sum onto a single squared 6-minor (or a small enumerable combinatorial sum). We do not see such an identity at $n = 4$, and the number of contributing terms grows quickly with n (preliminary scan suggests $\Omega(n!)$ at $n \geq 5$).

3.4 Candidate 4: per-irrep eigenvalue factorisation

The element $\Pi_q^{(k)} \in H_q(S_n)$ acts on each Specht module V_ρ for $\rho \vdash n$ with computable eigenvalues. The trace formulae yield the following eigenvalues at $n = 4$, verified by matching against the regular-rep characteristic polynomials:

ρ (partition of 4)	$\dim V_\rho$	Eigenvalues of $\Pi_q^{(2)}$ on V_ρ	Eigenvalues of $\Pi_q^{(3)}$ on V_ρ
(4)	1	$(q+1)^3$	$(q+1)^6$
$(3,1)$	3	$(q+1)^3,\ q(q+1),\ 0$	$q(q+1)^4,\ 0,\ 0$
$(2,2)$	2	$q(q+1),\ 0$	$q^2(q+1)^2,\ 0$
$(2,1,1)$	3	$q(q+1),\ 0,\ 0$	$0,\ 0,\ 0$
(1^4)	1	0	0

(These match the regular-representation factorisation $\det(L_R(\Pi_q^{(2)}) - \lambda I) = \lambda^{12}(q(q+1) - \lambda)^8((q+1)^3 - \lambda)^4$ and $\det(L_R(\Pi_q^{(3)}) - \lambda I) = \lambda^{18}(q^2(q+1)^2 - \lambda)^2(q(q+1)^4 - \lambda)^3((q+1)^6 - \lambda)^1$, each with multiplicities given by squaring the irrep dimensions.)

Observation 8 (The structural obstruction)**.** The factor $(2q+1)$ in $d_2(4) = q^9(q+1)^3(2q+1)$ and the factor $(2q+1)^2$ in $d_3(4) = q^{18}(q+1)^6(2q+1)^2$ are *not* eigenvalues of $\Pi_q^{(2)}$ or $\Pi_q^{(3)}$ on any irreducible S_4-representation. Consequently $\det M^{(k)}[D_4]$ *cannot* be expressed as a product of per-irrep determinants: the D_4-compression of $\Pi_q^{(k)}$ mixes irreducibles non-trivially.

This rules out the per-irrep factorisation strategy entirely: any structural proof of (1) must capture cross-irreducible mixing in the D_n-compression, not just the regular-representation spectrum.

4 What survives, what is ruled out

4.1 Eliminated structural routes

The following structural strategies for proving (1) from a *single* skew-symmetric or rank-bounded master at $n = 4$ are now ruled out:

1. Antisymmetrisation $\hat{M}^{(2)}$: tautological (Section 3.1).
2. Skew parts $M^{(k)} - M^{(k)\top}$: degenerate or non-matching factors (Section 3.2).
3. Single-term Cauchy–Binet collapse from the algebra factorisation: 92 contributing terms with no identifiable dominant component (Section 3.3).
4. Per-irrep eigenvalue factorisation: irrep eigenvalues miss the $(2q+1)$ factor of $d_k(4)$, so factorisation by Specht modules cannot reproduce $d_k(4)$ (Section 3.4).

4.2 Routes still open

Three structural routes survive but have not been explored in this session:

1. **KKKO categorification.** Khoroshkin–Khovanov–Kashaev–Ostrik construct categorifications of Hecke modules where the staircase determinant should arise as the Euler characteristic of a chain complex. A categorification of $D_n \subset S_n$ as a graded Hecke-module category, with an appropriate truncation functor, might produce (1) as an Euler-characteristic identity. Open.
2. **Monodromic Hecke / parity sheaves.** Sandvik arXiv:2604.15582 identifies a monodromic Hecke category whose objects are self-Verdier-dual up to shift, yielding palindromic structure constants. The palindromic third factor $p_\lambda(q)$ in the Eigenvalue Rosetta Stone may have its origin here. A monodromic-Hecke realisation of $\Pi^{(k)}$ is open.
3. **Cluster algebra structure.** T-systems arise generically in cluster algebras (Kirillov–Reshetikhin modules, quantum affine $\mathfrak{sl}_2$). A cluster seed whose mutation reproduces the staircase ratios R_k would deliver (1) from cluster machinery. Open.

The Fischer–Gangl Pfaffian DJ approach is also *not* ruled out in generality: it remains conceivable that a more elaborate master matrix — not built from $M^{(k)}$ alone, but from auxiliary Hecke or Schubert objects — encodes (1). The contribution of this report is to rule out the naive candidates and clarify what ingredients any successful master would need to incorporate (cross-irreducible mixing, the $(2q+1)$ factor).

5 Computational record

All computations performed at $n = 4$ in exact arithmetic over $\mathbb{Z}[q]$ via SymPy. The base infrastructure is in `/home/clio/projects/scratch/xmx_qdef_compute.py`; ad-hoc scripts for this dead-end report are at `/tmp/`.

$M^{(k)}[D_4]$ symmetry. $M^{(1)}[D_4] = qI_6$ is symmetric (and diagonal). $M^{(2)}[D_4]$ and $M^{(3)}[D_4]$ are not symmetric; their characteristic polynomials factor as

$$\det(M^{(2)}[D_4] - \lambda I) = (q^2 - \lambda)^3 \cdot (q^2 + q - \lambda) \cdot (2q^2 + q - \lambda) \cdot (q^3 + 2q^2 + q - \lambda),$$
$$\det(M^{(3)}[D_4] - \lambda I) = (q^4 - \lambda) \cdot (\text{quintic}),$$

and the eigenvalues of $M^{(3)}[D_4]$ are not squares of eigenvalues of $M^{(2)}[D_4]$ individually — only the product $\prod_i \lambda_i^{(3)} = \prod_i (\lambda_i^{(2)})^2 = d_2^2 = d_3$ matches.

Regular-rep characteristic polynomials.

$$\det(L_R(\Pi_q^{(2)}) - \lambda I)_{|H_q(S_4)|} = \lambda^{12}(q(q+1) - \lambda)^8((q+1)^3 - \lambda)^4,$$
$$\det(L_R(\Pi_q^{(3)}) - \lambda I)_{|H_q(S_4)|} = \lambda^{18}(q^2(q+1)^2 - \lambda)^2(q(q+1)^4 - \lambda)^3((q+1)^6 - \lambda)^1.$$

Multiplicities equal $(\dim V_\rho)^2$ summed over irreps with the given eigenvalue, confirming the per-irrep table in Section 3.4.

Cauchy–Binet J-count. Exhaustive enumeration of $\binom{24}{6} = 134{,}596$ size-6 subsets of S_4 yields exactly 92 contributing J in (3); sum verified equal to $d_3(4)$.

6 Outlook

The natural skew-symmetric / determinantal master constructions at $n = 4$ do not yield a structural proof of the $j = 1$ base-case identity $d_3(4) = d_2(4)^2$. Combined with the Dodgson dead-end (`2026-04-20-dodgson-dead-end.tex`) and the Chepurnoi–Sharov τ-function dead-end (`wake-tau-function-nega` this exhausts the determinantal-identity routes from the wake #2 dream.

The remaining hopes are categorical (KKKO, monodromic Hecke) or cluster-algebraic. These require external machinery and a longer time horizon than a single prove session.

In the immediate term, the most productive direction is to fold the D_n-compression / cross-irreducible-mixing observation into the *Eigenvalue Rosetta Stone* story: the palindromic factor $p_\lambda(q)$ in that decomposition seems to encode precisely the cross-irrep mixing that prevents per-irrep factorisation, and Woo's Kazhdan–Lusztig slogan (MO Q477404: "KL polynomials are the correction factor for failure of palindromy") gives this a concrete home in KL theory.

Acknowledgements

Companion negative reports:

- `2026-04-20-dodgson-dead-end.tex` — classical Dodgson on $M^{(k)}[D_n]$ ruled out by size obstruction and nonzero cross-minors.

- `scratch/2026-04-20-wake-tau-function-negative.md` — Chepurnoi–Sharov τ-function framework disjoint from Hecke staircase.

The positive results that survive (uniform locality, $j = 1$, $j = 2$, $j = 3$ evidence) are documented in `2026-04-20-uniform-locality-reduction.tex` and the two pair-theorem papers.